Ocean and Coastal Management 204 (2021) 105475

Contents lists available at ScienceDirect

Ocean and Coastal Management

journal homepage: http://www.elsevier.com/locate/ocecoaman

An uncommon guest is masked in fisheries landings: The case of the endemic sub-population of red spiny lobster Panulirus penicillatus (Decapoda: Palinuridae) in Costa Rica

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ARTICLE INFO

Keywords:
Costa Rica
Site-occupancy
Dive-based bentho-demersal fishery
Eastern Pacific ocean
Panulirus penicillatus

ABSTRACT

Fishery landings composition in tropical developing countries is formed of high diversity of species and some of them are not well-reported in landings or national fisheries statistics. This is particularly the case of the endemic red spiny lobster *Panulirus penicillatus* (Olivier, 1791) on the Pacific coast of Costa Rica. The red spiny lobster is a circumtropical species which has the most extensive global distribution of all the species of spiny lobsters. In order to fill the gap of knowledge regarding the spatial distribution of this species in the eastern Pacific Ocean and highlight its importance to fishing communities, this study focused on data collected from a dive-based bentho-demersal fishery on the northern Pacific coast of Costa Rica. Environmental and fishing effort data were recorded as an input for fitting site-occupancy models to get estimates of occurrence and probability of detection. The most parsimonious model structure for detection suggested that the covariates of diving time and water column visibility had an important effect on detection probability. For the occupancy part of the model, the depth gradient had the strongest effect, with high probability of lobster encounters in shallow rocky reefs near the coastline. These findings point towards an enlargement of the area of occurrence of the endemic sub-population of *P. penicillatus* to which sampled animals belong. Further, it is discussed potential ecological and fishery factors that explain the low occurrence of the red spiny lobster at this restricted area of the eastern Pacific continental shelf. As a result, a set of research guidelines and management recommendations are suggested for promoting the conservation and sustainable use of this endemic species within the region.

1. Introduction

The red spiny lobster, *Panulirus penicillatus* (Olivier, 1791), is the most widely distributed of the spiny lobsters, ranging throughout the Red Sea, Indo-Pacific and eastern tropical Pacific islands, including the Galápagos archipelago where it is found around most islands and islets, inhabiting the shallow rocky subtidal zone (Hickman and Zimmerman 2000), and Mexico (Briones and Lozano, 1982; Flores-Campana and Perez-Gonzalez, 1991). The species is often found in groups of more than 20 individuals of different sizes in sub-merged caves and lava tunnels where it demonstrates gregarious characteristics.

Two spiny lobster species (*Panulirus*) inhabit coastal waters on Costa Rica's Pacific coast. The most common of these species is the green lobster *Panulirus gracilis*, commonly present in the commercial landings from the country's small-scale fisheries (Naranjo, 2009). The least common of the two species, *P. penicillatus*, demonstrates off-shore restricted distribution around Coco's Island. Vargas and Cortés (1999) vaguely reported the presence of this species on the Costa Rican Pacific coast. However, the authors lack specific geographic information regarding its whereabouts and commercial use that cause uncertainty about its current spatial distribution and its historical utilization as a fishery resource.

In order to update the ecology and fishery knowledge of this species in the eastern Pacific Ocean, this study focused on data collected from a dive-based bentho-demersal fishery on the northern Pacific coast of Costa Rica in order to enhance the knowledge of marine biodiversity within the region and its importance to fishing communities.

2. Materials and methods

2.1. Data collection

Data were obtained from the commercial dive-based small-scale fishery activity (hookah and free diving) in Playa Lagarto (10° 07′,23"N -

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https://doi.org/10.1016/j.ocecoaman.2020.105475
Received 21 July 2020; Received in revised form 23 November 2020; Accepted 24 November 2020
Available online 3 December 2020

85° 47′,97"W) on Costa Rica's northern Pacific coast (Fig. 1). The red spiny lobster catches were identified through landings observations and on-board observations with fishers during the 2007–2008 and 2011–2012 fishing seasons (786 immersions or dives). From the aforementioned dataset, we used data on boats and free divers (who do not use boats to reach fishing sites) that visited fishing sites three times where red lobsters have been caught, totaling about 120 dives at 40 sites. The fishing trips that visited other fishing sites were excluded from the analysis. This helped to take away a disproportionate number of absences and reduce the effect of zero-inflation.

The rocky reef area comprised a fraction (5,4%) of the entire spatial distribution of fishing effort (132,2 km^2) (see Fig. 2a). Data recorded onboard boats during fishing trips included the following variables: depth of the fishing ground (m); diving time (hours); visibility/water clarity (recorded by Secchi disk); and location of the fishing areas (using a GPS, Garmin®).

Traditional Ecological Knowledge (TEK) of experienced divers was captured through interviews and onboard informal conversations. Their TEK provided additional information regarding the common names of fishing grounds, features of the fishing sites, and previous encounters with the species at temporal and spatial scales.

2.2. Site-occupancy model

The *Unmarked* package (Fiske and Chandler 2011) for the free statistical software R (R Development Core Team, 2014) was used for fitting ten reduced versions of site-occupancy models. The *occu* function fits the standard occupancy model based on zero-inflated binomial models (MacKenzie, 2006). It makes a formal distinction between the true occupancy state (z) and the observed occupancy state (y):

Detection probability: $p = \Pr(y_i = 1 | z_i = 1)$

Occurrence probability: $\phi = \Pr(z_i = 1)$

The occupancy state (z_i) process of site i is modeled as:

$$z_i \sim Bernoulli(\phi_i)$$

Here, z_i are unobservable latent variables or random effects and ϕ vary in response to spatial covariates (Royle and Dorazio, 2008).

The observation process is modeled as:

$$y_{ij} | z_i \sim Bernoulli(z_i p_{ij})$$

It is expressed the Bernoulli success probability as the product $z_i p_{ij}$. This implies that if the species is not present at a site i, so that $z_i = 0$, the y is a fixed 0. Otherwise, y_i is a Bernoulli trial with parameter p (Royle and Dorazio, 2008).

More generally, the *occu* function fits a hierarchical model which comprises two Bernoulli components expressed as two logistic regression models.

The effect of environmental conditions (visibility) and effort units (effective time) on *P. penicillatus* detection was modelled as a logistic function at 40 sites from Manzanillo and Frijolar rocky reef barriers, where the lobsters were caught during fishing seasons. Thus, we obtained a fishing pattern of detection and non-detection in temporal visits to fishing sites, which allows us to estimate true occurrence (ψ), free from distorted effects of imperfect detection (p) (MacKenzie, 2006; Kéry et al., 2010). The true occurrence is interpreted as the proportion of sites occupied by a species. Furthermore, the influence of site properties (depth) and fishing methods (hookah and free diving) were modelled under the assumption of perfect detection. We expected high probability of occurrence in shallow waters which is consistent with *P. penicillatus* habitat preferences. We also assumed that *P. penicillatus* might have high probability of occurrence for free divers because they work more frequently in shallower waters, less than 20 m depth.

The most complex model considered for occupancy was:

$$logit(\psi_i) = \alpha_0 + \alpha_1 \times fishing\ method_i + \alpha_2 \times depth_i$$

Here, ψ_i is the probability of site i being occupied, $\alpha_0$is the logit-linear intercept for the probability of occurrence of *P. penicillatus*, $\alpha_1$is the difference in probability of occurrence between Hookah and Free diving and α_2 denotes the depth effect in site i.

For detection, the most complex model considered was:

$$logit(p_{ij}) = \beta_0 + \beta_1 \times diving\ time_{ij} + \beta_2 \times visibility_{ij}$$

where, p_{ij} is the probability of detecting the species at site i during visit j, β_0 is the logit-linear intercept, β_1 is the effect of diving time in site i, during visit j and β_2 is the effect of visibility in the water column at site i

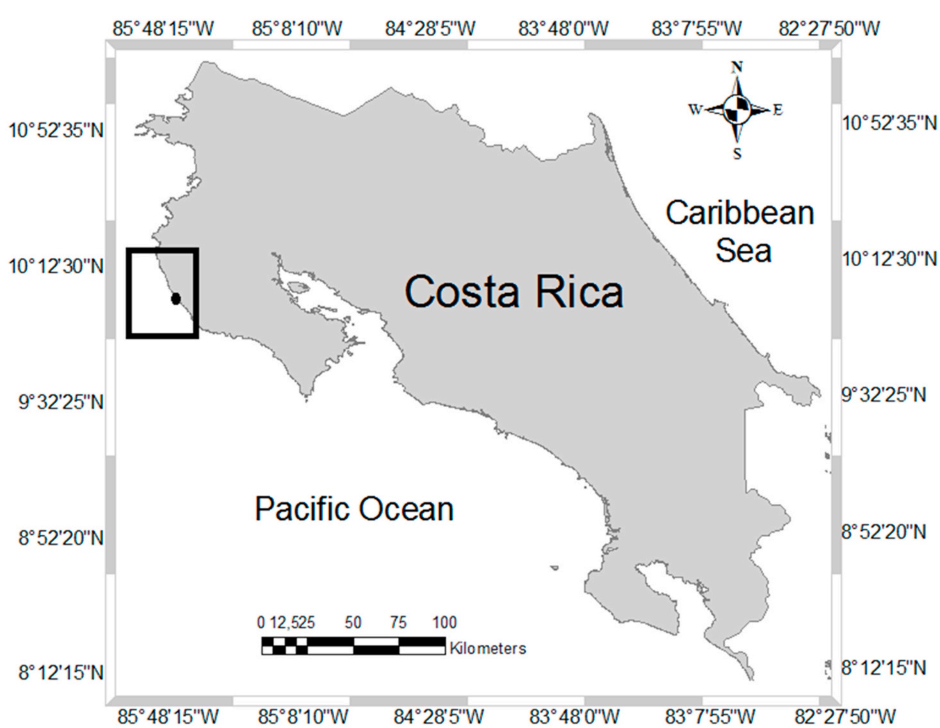

Fig. 1. Map showing the study area and Playa Lagarto's fishing port (black dot).

during visit *j*.

Model selection entailed the procedure suggested by Burnham and Anderson (2002), in which the most parsimonious model structure is identified given maximized log-likelihood and on the basis of a second order Akaike information criterion (AICc) to correct for bias. Moreover, models were ranked using the AICc differences and AICc weights as inferences for model comparison (Burnham and Anderson 1998). A parametric bootstrap analysis was conducted for assessing the goodness-of-fit of the best model following the procedure described by MacKenzie and Bailey (2004).

Predicted probability values for the best model were mapped using the Spatial Analyst extension of the Geographic Information System program ArcGIS 9.3. The procedure suggested by Riolo (2006) was used to produce density surfaces using a kernel density function, allowing for the calculation and visualization of discrete characteristics over space. In this case, the density was expressed as the probability of detection at each dive site (per km^2).

3. Results

A total of two female and four male red lobsters was observed and collected during the study period. Carapace length (CL) of female *P. penicillatus* ranged from 53 to 75.3 mm, and males displayed a range from 60 to 71.5 mm. Weight varied according to CL size and ovigerous condition. All individuals were collected in areas dominated by surf zones and rocky reefs from Manzanillo and Frijolar.

The most parsimonious model structure (see Table 1, model 2) for detection suggested that the covariates of dive time and water column visibility had an important effect on detection probability. Based on the parametric bootstrap, this model fitted the data well ($p = 0.02$). There was a constant effect of depth for all occupancy models. In addition, when the fishing method effect was dropped, the decrease of AICc was stronger than depth for the occupancy models, which means that both fishing methods were deployed in same fishing grounds along their depth gradient.

The latter pattern is clearly explained by the spatial distribution of *P. penicillatus* based on estimates of the best site-occupancy model, in which predicted probability of detection is higher in shallow rocky reefs near the coastline (see Fig. 2).

4. Discussion

It was shown how fishery-dependent sampling can be used to determine occurrent-based distribution of species in contexts where historical fishery-dependent or fishery-independent data is rare or not available. The generality and simplicity of the sampling design yielded information about probability of occupancy and probability of detection that may provide the necessary feedback to guide monitoring programs and fishery policy in data-poor contexts.

Observations on catch landings during the sampling period did not report high amounts of *P. penicillatus*. Compared to the total catch volume of *P. gracilis*, the red spiny lobster represented only 0.98% of the total catch landed (Naranjo, 2014). Nonetheless, the fishing effort for both fishing methods entailed a broad spatial extent with multiple fishing sites along the coast (see Naranjo-Madrigal and van Putten, 2019), the occurrence of *P. penicillatus* was restricted to a reef barrier characterized by surf zones near the coast which is consistent with its preferred habitats.

Our findings are in accordance with previous studies wherein fishing effort increased (e.g., diving hours and number of divers or dives) for both hookah and free divers during the dry months when water visibility and thus catches were higher (Naranjo, 2014). Therefore, oceanographic conditions and diving time had an important effect on detection probability of *P. penicillatus*, especially in shallow rocky reefs near the coastline.

Based on these findings, the extent of occurrence (Gaston and Fuller,

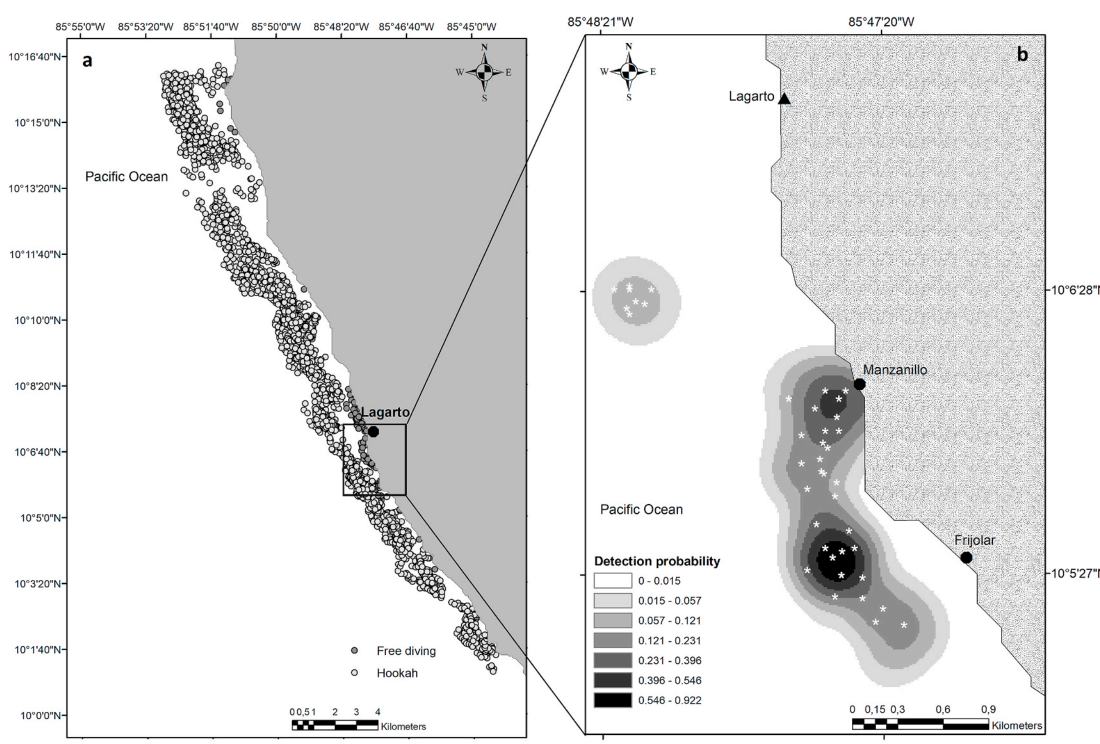

Fig. 2. Map showing diving sites of the entire spatial distribution of fishing effort (a). Diving sites where red lobsters *P. penicillatus* were caught and common names of the beaches in front the rocky reef barriers where fishing trips took place between 2007-2008 and 2011–2012 (b). Predicted detection probabilities of the red spiny lobster (*P. penicillatus*) on Costa Rica's northern Pacific coast. Estimates are based on the most plausible model (Model 2). The red spiny lobsters are less likely to be detected in sites further from the coastline. Black triangle depicts main fishing port and black dots show the name of the fishing sites. White asterisks depict diving sites.

Table 1
Results for fitting ten site-occupancy models to the detection of the red spiny lobster *P. penicillatus* on Costa Rica's northern Pacific coast between 2007-2008 and 2011–2012. Negative log-likelihood (NLogLik) and corrected Akaike Information Criteria (AICc) scores are shown for each model fitted. ΔAICc values below 2 support much weight of evidence in favor of being the best model. AICc weights (AICcWt) vary from 0 (no evidence) to 1 (evidence). Boldface shows the most parsimonious model.

	Occupancy Probability			Detection Probability					Model-selection descriptors	
	Intercept	Fishing method	Depth	Intercept	Diving time	Visibility	NLogLik	AICc	ΔAICc	AICcWt
Model 2	−15.3	–	2.0	−4.5	3.05	2.04	−15.42	42.61	0	0.43
Model 10		–	–	−4.3	2.99	1.94	−17.19	43.51	0.9	0.27
Model 1	0.4	9.23	–	−4.5	3.07	2.01	−16.21	44.17	1.56	0.19
Model 3	−13.9	5.10	1.82	−4.5	3.05	2.04	−14.72	45.39	2.78	0.11
Model 8	2	–	−0.09	−2.8	2.79	–	−24.15	57.44	14.83	0.00
Model 7	0.3	0.11	–	−2.9	2.86	–	−24.68	58.51	15.9	0.00
Model 9	1.5	1.49	−0.14	−2.8	2.84	–	−23.85	59.49	16.88	0.00
Model 5	1.1	–	−0.08	−1.4	–	1.84	−29.34	67.81	25.2	0.00
Model 4	−0.3	−0.04	–	−1.5	–	1.84	−29.95	69.05	26.44	0.00
Model 6	0.8	1.51	−0.13	−1.4	–	1.88	−28.92	69.61	27	0.00

2009) of *P. penicillatus* should be expanded, considering that these specimens are part of a phylogenetic linage of an endemic sub-population that was isolated by the East Pacific Barrier 1.5 Ma (Iacchei et al., 2016). According to Iacchei et al. (2016) and George (2005) the dark-reddish brown form of *P. penicillatus* is regionally restricted to the east Pacific at the Galapagos, Clipperton, and Revillagigedo Islands, and during its evolution it may have adapted larval retention behaviours similar to other Panuliridae to maintain local populations during the changing currents at the beginning of the Pleistocene.

In addition, the small area of occurrence of *P. penicillatus* on the north Pacific coast of Costa Rica could be an approximate portrayal of its current local abundance. To explain its low occurrence, it is necessary to understand potential mechanisms underlying colonization, extinction, and fish depletion dynamics.

4.1. Fishery interactions

The reconstructed TEK from older fishers' experiences (see Naranjo-Madrigal and van Putten, 2019) confirmed the presence of *P. penicillatus* in the region for more than 40 years, known as "Limonense" lobster referring to their red colored exoskeleton, similar to those caught (*Panulirus argus*) on the Caribbean coast in the province of Limon, Costa Rica.

Since then, *P. penicillatus* have been subjected to increasing exploitation rates similar to those suffered by the green lobster *P. gracilis* (Naranjo, 2011), given the fact that fishermen have never sorted out lobster species during catch landings and they were sold as the same species. Consequently, it could be deduced that the same exploitation patterns that leads to the overexploitation of *P. gracilis* may have gradually depleted the biomass of *P. penicillatus,* which partly explains the low occurrence registered in this study.

Chronic and intense semi-industrial or industrial bottom trawling (Pusceddu et al., 2014) is another cause that negatively impacts different habitats and life-stages of many benthic species, including *P. penicillatus*. The local semi-industrial trawl fleet in particular, has shown sequential overexploitation of target species and a high level of bycatch with subsequent spatial effort displacement from shallow to deep waters (Palacios, 2013). The trawl fleet regulatory framework has been characterized by the absence of ecologically-based spatial management plans with information about habitat structural and/or functional sensitivity to fishing trawling (e.g. see Bolam et al., 2014). Local fishermen have reported trawling nets lost at rocky reefs and clandestine trawling landings of demersal fish species (personal communication, April 12, 2007). For more than 6 decades, inefficient enforcement and unrestricted trawling operations may have been affecting indiscriminately important habitats for *P. penicillatus*, such as rocky reefs and corals (Bystrom et al., 2017; Lequeux et al., 2018).

4.2. Ecological interactions

In addition to historical fishery-dependent effects on *P. penicillatus* abundance, other potential macroecological processes could explain the low occurrence of the red spiny lobster on the northern Pacific coast of Costa Rica. It is well-known that the broad geographic distribution of marine benthic invertebrates may be caused by the dispersal of pelagic larvae by ocean currents (Hearn and Murillo, 2008). For example, Chow et al. (2011) identified, based on a study of phyllosoma larvae distribution, the Galapagos Islands as *P. penicillatus'* main habitat, with a lower abundance on the neighboring continental coast to the east. Briones-Fourzan et al. (2019) pointed out that periodic recruitment events may explain the presence of *P. meripurpuratus* along the Mexican Caribbean coast. However, the biogeographic configurations of the sea bed (e.g. shallow seamounts, knolls, hills, islets) could play a crucial role in the distribution of this species along the coast of Central America. Sea bottom configurations (e.g. shallow seamounts, see Du Preez et al., 2016) may work as connectors that allow the colonization of adults and settlement of larvae at various points along the continental shelf.

There is strong evidence of the existence of source-sink relationships between regional breeding of spiny lobster stocks and puerulus settlement (Lipcius et al., 2001; Feng et al., 2011), which supports the assumption of a potential unidentified source zones which might serve as habitat reservoirs for recruitment, and in turn as transit areas for potential migration and colonization patterns of this species throughout the eastern Pacific Ocean. This evidence points out the importance of considering benthonic species spatial-distribution across potential sources zones as an input to transboundary initiatives for habitat protection and biodiversity conservation; conversely, regional efforts of marine protection has been focused solely on pelagic flagship species (e.g. turtles, sharks) migratory patterns and their spatial distribution (Amorocho et al., 2012; Nalesso et al., 2019).

5. Recommendations

The identification of the stock size and spatial range of spiny lobsters such as *P. penicillatus* is important as an input for conservation and fishery management. For this reason, fishery-dependent and independent research to determine potential sources zones, and whether these animals belong to a migratory or resident population is highly recommended. Also, this research highlights the need to study and identify the natural factors that may influence the dispersal of *P. penicillatus* phyllosoma larvae and recruitment variability in the region. This information could guide further spatial-explicit protection measures and help to develop harvest strategies for its sustainable management.

Furthermore, it is urgent to remove human induces factors which are perturbing (e.g. semi-industrial fishing trawling) population displacement along *P. penicillatus* broad range of distribution. Fishing mortality should be stopped by means of a ban on *P. penicillatus* catch with the aim to increase recolonization and hamper potential process of local extinction. In the long-term, these measures could lead to habitat improvement and abundance recovery of *P. penicillatus* within the eastern Pacific Ocean.

Declaration of competing interest

The author declares that he has not known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The author thanks the anonymous referees for their constructive comments. I thank the fishers of Playa Lagarto, who made this study possible: Luis Calderon, Yehudy Calderon, Junior Espinoza, Leiter, Greivin, Pelón, Richard, Ata, Loro. I also thank the middlemen and all of those who collaborated in this study in one way or another. I warmly thank Andrew Bystrom for his remarkable language editing work. I also thank Sarah Converse at University of Washington and Alex Hearn at Universidad San Francisco de Quito for their valuable comments on the manuscript. Financial support from ALNA S.A (Grant number: UN-039) is acknowledged by the author.

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